

Project: AI Data Cleaning & EDA Agent

<<<<<< HEAD ■ AI DATA CLEANING AND EDA AGENT

An intelligent data preprocessing and exploratory data analysis tool powered by AI for machine learning workflows. ****Now supports datasets up to 2GB with intelligent memory optimization!****

■ FEATURES

■ ****INTELLIGENT DATA CLEANING****

• ****Large Dataset Support****: Handle datasets up to 2GB with automatic optimization • Automated missing value detection and imputation • Outlier identification using statistical and ML methods • Data type optimization and validation (up to 90% memory reduction) • Duplicate detection with smart merge suggestions • Inconsistent data pattern recognition • Memory-efficient chunk processing for very large files

■ ****AI-POWERED EDA****

• Automated descriptive statistics generation • Smart visualization recommendations ■ AI DATA CLEANING AND EDA AGENT

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■ ****AI-POWERED EDA****

• Automated descriptive statistics generation • Smart visualization recommendations • Pattern detection and anomaly identification • Feature importance analysis • Correlation matrix with insights • Distribution analysis with recommendations

■ ****ADVANCED PREPROCESSING****

• Automated feature engineering suggestions • Text preprocessing for NLP tasks • Image preprocessing for computer vision • Time series preprocessing and feature extraction • Categorical encoding optimization

■ ****COMPLETE ML PIPELINE****

• Automated model selection based on data characteristics • Multi-model training with hyperparameter optimization • Cross-validation and performance evaluation • Feature importance analysis and selection • Auto-ML pipeline for end-to-end workflow • Model comparison and recommendation system

■ ****INTERACTIVE DASHBOARD****

• Real-time data quality metrics • Interactive visualizations • Automated insights and recommendations • Complete ML workflow interface • Export capabilities for reports • Integration with popular ML frameworks

■■ ****TECH STACK****

• ****Core****: Python 3.9+ • ****Data Processing****: Pandas, NumPy, Polars, Dask • ****Large Dataset Support****: PyArrow, FastParquet, Memory Optimization • ****Machine Learning****: Scikit-learn, XGBoost, LightGBM • ****Visualization****: Plotly, Seaborn, Matplotlib • ****Web Interface****: Streamlit • ****Performance Monitoring****: psutil, Memory Management

■ ****QUICK START****

```
```bash CLONE THE REPOSITORY
```

```
git clone https://github.com/yourusername/ai-data-cleaning-agent cd ai-data-cleaning-agent
```

## INSTALL DEPENDENCIES

```
pip install -r requirements.txt
```

## RUN THE APPLICATION

```
streamlit run app.py ```
```

## ■ **\*\*USAGE\*\***

## WEB INTERFACE (RECOMMENDED)

```
```bash streamlit run app.py ```
```

PYTHON API

```
```python from src.data_cleaning_agent import DataCleaningAgent from src.preprocessing_agent
import
PreprocessingAgent from src.ml_model_agent import MLModelAgent
```

## COMPLETE AI-POWERED WORKFLOW

```
cleaning_agent = DataCleaningAgent() preprocessing_agent = PreprocessingAgent() ml_agent =
MLModelAgent()
```

## LOAD AND CLEAN DATA

```
cleaned_data = cleaning_agent.auto_clean_data(df)
```

## INTELLIGENT PREPROCESSING

```
preprocessed_data = preprocessing_agent.auto_preprocess(cleaned_data, target_column='target')
```

## AUTO-ML PIPELINE

```
ml_results = ml_agent.auto_ml_pipeline(preprocessed_data['X_train'],
preprocessed_data['y_train']) ```
```

### ■ **\*\*PERFECT FOR\*\***

• Amazon ML Challenges • Kaggle competitions • Data science projects • ML model preprocessing • Automated data quality assessment

### ■ **\*\*CONTRIBUTING\*\***

Contributions welcome! Please read our contributing guidelines.

### ■ **\*\*LICENSE\*\***

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--- **\*\*Built for the future of automated data preprocessing! ■\*\***